

Complete Mushroom Farming Guide

Satna, Madhya Pradesh — Year-Round Plan with Bamboo Hut & Room Setup

Why Satna is Ideal for Mushroom Farming

Satna has all three things mushroom farming needs — hot summers (for milky mushroom), cool dry winters (for oyster and button), and monsoon humidity (for paddy straw mushroom). With bamboo and land already available, your setup cost is very low compared to most farmers.

- Hot summers 35–45°C → perfect for Milky mushroom (March–September)
- Cool winters 10–22°C → ideal for Oyster & Button (October–February)
- Monsoon humidity → Paddy straw mushroom thrives (June–September)
- Bamboo + land available → hut cost as low as Rs 8,000–15,000
- Government subsidy available from KVK Satna & ATMA office

1. Year-Round Growing Calendar — Which Mushroom, Which Season

Mushroom	Best Season	Temperature	Difficulty	Price (Fresh)
Oyster (Dhingri)	Oct – Mar	18–28°C	Easy	Rs 80–120/kg
Button (Safed)	Oct – Feb	14–18°C	Medium	Rs 100–150/kg
Milky (Dudhi)	Mar – Sep	28–38°C	Easy–Med	Rs 60–100/kg
Paddy Straw	Jun – Sep	28–35°C	Easy	Rs 40–70/kg
Shiitake	Nov – Feb	15–22°C	Hard	Rs 250–400/kg
Reishi (Ganoderma)	Apr – Aug	25–32°C	Medium	Rs 500–1000/kg dry

Strategy for continuous income:

- Oct–Feb: Grow Oyster + Button (main income months)
- Mar–Sep: Switch to Milky mushroom + Paddy straw
- All year: Add Reishi / Shiitake on sawdust logs as premium crop

2. Spawn (Seed) — Types, Sources & Storage

Spawn is the 'seed' of mushroom farming — mycelium grown on a carrier material. Quality spawn = healthy crop. Never compromise on spawn quality.

Spawn Type	Carrier Material	Best For	Shelf Life
Grain spawn	Sorghum / wheat grain	Oyster, Milky	45 days @ 4°C

Spawn Type	Carrier Material	Best For	Shelf Life
Sawdust spawn	Hardwood sawdust	Shiitake, Reishi	60 days @ 4°C
Straw spawn	Paddy / wheat straw	Paddy straw, Milky	30 days @ 4°C
Compost spawn	Composted manure	Button mushroom	45 days @ 4°C

Where to buy spawn in MP:

- JNKVV Jabalpur — government lab, reliable quality (Rs 15–25/kg)
- KVK Satna / Rewa — sometimes free or subsidised for registered farmers
- Local agri shops in Satna city — buy only sealed, dated packets
- IARI Delhi — postal delivery available (Rs 30–40/kg)

Spawn rate: use 3–5% of dry substrate weight. For 100 kg dry straw = 3–5 kg spawn. Store in refrigerator at 4°C. Never use spawn stored at room temperature in summer — it will be dead.

3. Substrate (Bedding) Mix — Complete Recipes

3A. For Oyster Mushroom and Milky Mushroom

Ingredient	Proportion	Purpose
Wheat straw (gehu bhusa)	70%	Main carbon source
Paddy straw	20%	Bulk, structure
Lime (chuna)	1%	pH control (raise to 7–8)
Gypsum	1%	Calcium, structure
Water	To 60–65% moisture	Hydration

Preparation steps:

- Step 1: Chop straw into 3–5 cm pieces
- Step 2: Soak in water for 12 hours
- Step 3: Pasteurise — boil in drums or pour boiling water, maintain 80°C for 60–90 minutes
- Step 4: Drain and spread on clean plastic sheet. Cool to room temperature.
- Step 5: Test moisture — squeeze handful, only 2–3 drops should drip out

3B. For Button Mushroom (Agaricus)

Ingredient	Proportion	Notes
Wheat straw	50%	Chopped fine
Cow / horse dung (dried)	30%	Essential for nutrition
Urea	0.5%	Nitrogen source
Gypsum (CaCO ₃)	2%	pH + structure
Superphosphate	1%	Phosphorus

Button compost needs 21–28 day fermentation in 2 phases. Phase I: outdoor heap, turn every 3–4 days. Phase II: pasteurise at 55–60°C for 4 days. Casing layer: 1:1 red soil + coir dust (sterilised).

4. Bamboo Hut Setup — Your Land & Bamboo Used

A bamboo hut is the lowest cost, most practical structure for MP climate. It provides natural insulation, good ventilation, and can be built in 2–3 days with local labor.

Component	Specification / Material	Notes
Size (1 unit)	20 ft × 10 ft × 8 ft height	Holds 600–800 bags
Frame	Bamboo poles 3–4 inch dia, planted 2 ft deep	Space poles 4 ft apart
Roof	Bamboo + grass thatch OR tarpaulin	Add 50% shade net on top
Walls	Bamboo weave + black polythene inside	Polythene keeps humidity in
Shelves	4-tier bamboo shelves, 1.5 ft wide	Along both long walls
Door	Single bamboo door with plastic curtain	Keeps insects out
Ventilation	2 small windows + 1 exhaust hole at top	Open at night in summer
Flooring	Packed mud / brick	Easy to wash and clean

Hut cost estimate (using your own bamboo):

- Labor (2 workers, 3 days): Rs 2,400–3,600
- Polythene sheet (200 sqft): Rs 600–800
- Shade net (50%): Rs 1,200–1,800
- Tarpaulin for roof: Rs 800–1,200
- Miscellaneous (nails, wire, rope): Rs 400–600
- Total hut cost (bamboo from own land): Rs 5,400–8,000

2 huts (20×10 ft each) = 1,200–1,600 bags per batch. Run staggered batches for continuous weekly harvest.

5. Indoor Room Setup — If Growing Inside a Room

A closed room gives better control over temperature and humidity — ideal for button mushroom and for summer months when hut gets too hot.

Step 1	Whitewash all walls + ceiling. Spray 2% Bavistin (carbendazim) solution on all surfaces. Wait 48 hours, then air out.
Step 2	Install 4–5 tier shelves (bamboo or iron angle). 18-inch wide shelves. Along 3 walls.
Step 3	Exhaust fan on one wall (top position). Air inlet with muslin/net filter on opposite wall (bottom position).
Step 4	Mist sprayer or humidifier on timer — set to spray 2–3 times per day. Target: 80–90% humidity.
Step 5	Install thermometer + hygrometer (Rs 150–200 from agri shop). Monitor daily.
Step 6	In summer, add a wet gunny bag on the exhaust inlet side or use an evaporative cooler (Rs 3,000–5,000).

6. Temperature Management — Satna Month-by-Month

Month	Outside Temp	Crop to Grow	Action Needed
Oct–Feb	10–22°C nights	Oyster, Button, Shiitake	No cooling needed. Ideal season.
March	28–35°C	Switch to Milky mushroom	Start closing hut at midday
Apr–May	38–45°C days	Milky mushroom, Reishi logs	Wet gunny on walls, shade net
Jun–Sep	Humid 28–35°C	Milky, Paddy straw, Reishi	Increase ventilation, open nights
Summer (any)	Above 35°C	Any crop indoors	Evaporative cooler in room

7. Step-by-Step Growing Process (Oyster Mushroom Example)

1. Prepare substrate

Chop straw (3–5 cm). Soak 12 hrs. Pasteurise at 80°C for 60–90 min. Drain. Cool to room temp. Check moisture (60–65%).

2. Fill and spawn bags

Use HDPE bags (45×60 cm). Layer: 5 cm straw → spawn layer → repeat 4–5 times. Tie top with rubber band. Make 4–5 small holes on sides for air.

3. Spawn run (incubation)

Keep bags in dark room at 22–28°C. Do NOT water yet. Mycelium will spread — bag turns white. Takes 10–18 days. If any green/black mold appears, discard that bag.

4. Fruiting initiation

Once bag is fully white, cut 4–6 cross slits (2 cm each) on sides. Or remove top of bag. Shift to fruiting room/hut. Begin watering 2–3 times daily.

5. Pinning and harvest

Small mushroom pins appear in 3–5 days. Harvest when caps are fully open but before edges curl upward. Twist and pull — do not use knife. Each bag gives 3–4 flushes over 60–70 days.

6. After harvest

Rest bag 5–7 days between flushes. Keep humidity high. After final flush, spent substrate = excellent compost. Sell at Rs 5–8/kg to farmers.

Expected yield: 800g to 1.2 kg fresh mushroom per kg dry straw (biological efficiency 80–120%).

8. Profit Analysis — One 20x10 ft Bamboo Hut

Item	Details / Amount
Bags per batch	600 bags
Dry straw required	~300 kg

Spawn required	15 kg @ Rs 25/kg = Rs 375
Total input cost (per batch)	Rs 4,000–6,000 (straw, spawn, lime, bags, labor)
Expected yield per batch	400–600 kg fresh oyster mushroom
Revenue @ Rs 80–100/kg	Rs 32,000–60,000
Net profit per batch	Rs 26,000–54,000
Batches per year	4–5 (oyster Oct–Mar + milky Apr–Sep, staggered)
Annual net profit (1 hut)	Rs 1,00,000 to Rs 1,80,000
2 huts (double)	Rs 2,00,000 to Rs 3,60,000

9. Additional Income Streams Beyond Fresh Sales

Income Stream	How to Do It	Potential Revenue
Dried mushroom	Sun-dry or food dryer. 10 kg fresh = 1 kg dry.	Rs 600–800/kg dried
Mushroom pickle	Oyster mushroom pickle, home recipe.	Rs 200–400/500g jar
Mushroom powder	Dry + grind. Sell to ayurvedic shops.	Rs 300–500/100g
Own spawn production	Learn at KVK. Sell to local farmers.	Rs 25–40/kg spawn
Farmer training	Charge Rs 500–1,000/person for hands-on demo.	Rs 5,000–20,000/month
Spent substrate (compost)	Sell after final flush to vegetable farmers.	Rs 5–8/kg
Reishi medicinal	Sell dried caps to ayurvedic dealers.	Rs 500–1,000/kg dry

10. Market Channels from Satna

Local / direct selling:

- Sabzi mandi — Satna city, Rewa, Maihar (daily fresh sales)
- Hotels, dhabas, restaurants — approach directly with samples
- Local grocery chains and supermarkets in Satna
- Vegetable vendors — supply on commission basis

Online / processed:

- Amazon Fresh, BigBasket — for dried mushroom and mushroom powder
- IndiaMart.com — list your produce for bulk buyers
- WhatsApp community selling — create a local customer group

Institutional:

- NDDDB (dairy cooperative) and FPOs (Farmer Producer Organisations) in MP
- Ayurvedic medicine stores for reishi and shiitake powder
- School and hospital canteens — supply contract, stable demand

11. Pros and Cons

Pros	Cons / Challenges
Low land requirement (1–2 rooms or small hut)	Contamination risk if hygiene is poor (main failure reason)
Bamboo reduces setup cost massively	Summer heat management for oyster mushroom
Year-round income with different varieties	Button mushroom needs more skill and 21-day compost prep
Quick return — first harvest in 30–45 days	Fresh mushroom is perishable, needs daily selling
No heavy machinery needed	Market price can fluctuate locally in peak season
Government subsidy available	Requires daily attention — watering, monitoring, harvesting
Spent substrate = additional compost income	Learning curve for first 2 batches is steep

12. Government Support — Satna, MP

- Contact KVK Satna (Krishi Vigyan Kendra) — free training, subsidised spawn, technical guidance
- ATMA Office Satna — register as mushroom farmer, get scheme benefits
- National Horticulture Mission (NHM) — 25–50% subsidy on shed construction
- PMKVY (Pradhan Mantri Kaushal Vikas Yojana) — free skill training including mushroom cultivation
- MP Horticulture Department — ask for 'mushroom cluster development scheme'
- Mukhyamantri Krishak Udyami Yojana — up to Rs 10 lakh loan for agri enterprise at subsidised rate

13. Your Action Plan — Month by Month

Timeframe	What To Do
Month 1 (now)	Visit KVK Satna. Register. Collect free training dates. Buy 5 kg grain spawn. Build 1 hut (10x10 ft). Prepare 1
Month 2	First harvest! Sell locally. Learn from any contamination. Start second batch of 200 bags.
Month 3–4	Scale to 400–600 bags in 1 hut. Approach 2–3 restaurants/hotels with samples. Start drying 10–15% of harve
Month 5–6	Build second hut. Start Milky mushroom batch for summer. Open IndiaMart listing for dried mushroom.
Year 2	Add Shiitake on sawdust logs. Start own spawn production. Offer training to other farmers (Rs 500–1,000/hea

Prepared for Satna, Madhya Pradesh mushroom farming. Always verify spawn quality, maintain strict hygiene, and consult KVK Satna for local guidance. Best of luck with your mushroom farming enterprise!